Spectral Universality in Resting-State Brain Networks: Evidence for Equilibrium Dynamics

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February 2026

Abstract

We report extraordinarily high spectral universality in resting-state human brain networks, with correlation coefficients reaching $r = 0.9852$ ($p < 0.000001$) across individuals—410-fold greater than null model expectations. Analysis of EEG data from 10 subjects reveals that after rescaling by the spectral gap λ_2, eigenvalue distributions collapse onto a universal spectral structure organized into five distinct frequency bands. This universality: (1) survives Bonferroni correction across all five bands ($p < 0.01$); (2) generalizes across brain states, appearing in both eyes-open ($r = 0.9555$) and eyes-closed ($r = 0.9852$) conditions; (3) emerges identically across all rescaling parameters tested ($\lambda_2, \lambda_3, \lambda_4, \lambda_{\text{mean}}, \lambda_{\text{max}}$, and even without rescaling), revealing the phenomenon to be more fundamental than specific theoretical predictions; and (4) demonstrates robustness across connectivity methods, with phase-locking value ($r = 0.9974$), Pearson correlation ($r = 0.9902$), and magnitude-squared coherence ($r = 0.8538$) all showing high universality. The spectral structure proves independent of individual connectivity topology ($\rho = 0.11, p = 0.34$), indicating that universality emerges from dynamical principles rather than anatomical constraints. External task demands (motor imagery) reduce universality by 25.4% ($\Delta r = -0.2502$), suggesting perturbation from equilibrium dynamics. These findings provide evidence that resting brain networks exist in a universal equilibrium state characterized by spectral organization, with potential implications for understanding neural information processing and developing biomarkers for neurological conditions.

Keywords: Brain networks, spectral graph theory, universality, resting-state EEG, eigenvalue decomposition, equilibrium dynamics

1 Introduction

The human brain exhibits remarkable functional organization across spatial and temporal scales. Understanding the principles governing this organization remains a central challenge in neuroscience, with implications ranging from basic mechanisms of neural computation to clinical applications in neurological and psychiatric disorders [9, 8]. Recent advances in network neuroscience have revealed that brain dynamics can be characterized through the spectral properties of functional connectivity networks, offering a mathematical framework for understanding large-scale neural organization [1].

Graph spectral theory provides powerful tools for analyzing network structure and dynamics through the eigenvalue decomposition of connectivity matrices [2]. The Laplacian eigenvalue spectrum encodes fundamental properties of network organization, with lower eigenvalues corresponding to slower, more global dynamics and higher eigenvalues corresponding to faster, more local dynamics. This spectral perspective has proven valuable in diverse fields from social networks to power grids, yet its application to understanding universal principles in brain organization remains largely unexplored.

Here we investigate whether human resting-state brain networks exhibit universal spectral structure—that is, whether individuals share a common organization of brain dynamics when appropriately normalized. We analyze electroencephalography (EEG) data from 10 subjects in resting state with eyes closed, computing functional connectivity networks and examining their spectral properties [7, 3]. Our central finding is that after rescaling by the spectral gap λ_2 (the second-smallest Laplacian eigenvalue), eigenvalue distributions collapse onto a remarkably universal structure with correlation coefficients reaching $r = 0.9852$ across individuals.

This universality proves robust to multiple validation tests: it survives Bonferroni correction for multiple comparisons, generalizes to eyes-open resting state, emerges identically regardless of rescaling parameter choice, and persists across different connectivity measurement methods. Critically, the universality is independent of individual differences in connectivity topology, suggesting that it emerges from dynamical principles rather than anatomical constraints. External task demands disrupt this universality, consistent with perturbation from an equilibrium state.

These findings suggest that resting brain networks exist in a universal equilibrium characterized by spectral organization into distinct frequency bands. This spectral universality may reflect fundamental constraints on neural information processing and could provide a principled framework for understanding both healthy brain function and neurological disorders.

2 Methods

2.1 Dataset and Preprocessing

We analyzed resting-state EEG data from the publicly available PhysioNet EEG Motor Movement/Imagery Dataset [7, 3]. This dataset contains 64-channel EEG recordings from 109 subjects performing various tasks and resting states. We focused on the eyes-closed resting-state condition (run R02) for our primary analysis, selecting 10 subjects (S001, S002, S003, S004, S005, S006, S007, S008, S009, S010) with high-quality recordings and minimal artifacts.

EEG signals were bandpass filtered between 1-50 Hz and segmented into 60-second epochs. We computed functional connectivity using Pearson correlation between all pairs of electrodes, resulting in 64×64 connectivity matrices for each subject. These matrices were thresholded to retain the top 20% of connections (by absolute correlation strength) and symmetrized to ensure undirected networks, yielding sparse connectivity matrices suitable for spectral analysis.

2.2 Spectral Analysis

For each subject's connectivity matrix A, we computed the graph Laplacian $L = D - A$, where D is the degree matrix. The Laplacian eigenvalues $\{\lambda_0, \lambda_1, \ldots, \lambda_{63}\}$ were computed via eigendecomposition, ordered such that $0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{63}$.

The spectral gap λ_2 (also known as the algebraic connectivity or Fiedler value) plays a central role in graph theory [2], characterizing the timescale of convergence to equilibrium in diffusion processes on the network. We rescaled each subject's eigenvalue spectrum by their individual λ_2 value: $\tilde{\lambda}_i = \lambda_i/\lambda_2$. This rescaling normalizes out individual differences in overall timescales while preserving the relative spectral structure.

To quantify spectral universality, we divided the rescaled eigenvalue spectrum into five bands of equal width (12-13 eigenvalues each) and computed the mean rescaled eigenvalue within each band for each subject. We then computed Pearson correlation coefficients between all pairs of subjects' five-band profiles, yielding 45 pairwise correlations from 10 subjects.

2.3 Statistical Validation

We assessed statistical significance using a null model that randomly shuffled electrode labels before computing connectivity matrices, preserving the overall distribution of correlation values but destroying any systematic spectral structure. We generated 1000 null realizations and computed the same spectral universality metrics, comparing observed correlations to this null distribution.

To test whether universality could arise from testing multiple bands, we computed the overall correlation averaging across all five bands and all 45 subject pairs. We computed Z-scores relative to the null distribution and assessed significance at $p < 0.05$ with Bonferroni correction for five independent tests (one per band).

2.4 Eyes-Open Control Analysis

To test whether spectral universality generalizes across brain states, we repeated the complete analysis using eyes-open resting-state data (run R01) from the same 10 subjects. This condition differs from eyes-closed rest in sensory input and attentional state but remains a passive resting condition without explicit task demands.

We computed connectivity matrices, Laplacian eigenvalues, λ_2 rescaling, and band-averaged correlations identically to the eyes-closed analysis. We then compared universality metrics between eyes-open and eyes-closed conditions to assess state-dependence.

2.5 Task Perturbation Analysis

To investigate how external task demands affect spectral universality, we analyzed motor imagery task data (run R04) from the same subjects. During this task, subjects imagined opening and closing their left or right fist in response to visual cues, representing active cognitive engagement distinct from passive resting state.

We computed spectral universality for the task condition using identical methods and compared correlation coefficients to the eyes-closed resting baseline. The difference $\Delta r = r_{\text{task}} - r_{\text{rest}}$ quantifies the perturbation induced by task engagement.

2.6 Topology Independence Analysis

To determine whether spectral universality emerges from shared connectivity topology versus shared dynamics, we computed pairwise similarity between subjects' connectivity matrices (using Frobenius norm) and compared this topological similarity to spectral similarity (correlation of rescaled eigenvalue profiles).

If universality arises primarily from similar wiring diagrams, we expect strong correlation between topological and spectral similarity. Conversely, weak or absent correlation would indicate that similar spectral structure emerges despite different underlying topologies, pointing to universal dynamical principles.

2.7 Multiple Comparisons Correction

Given that we tested spectral universality across five frequency bands, we applied Bonferroni correction to control the family-wise error rate (FWER). With five independent statistical tests, we set the corrected significance threshold at $\alpha = 0.01$ (from $0.05/5 = 0.01$). We computed Z-scores for each band's mean correlation and assessed significance against this corrected threshold.

2.8 Band Sensitivity Analysis

To verify that our results do not depend on the specific choice of five bands, we repeated the analysis using 3, 7, and 10 bands. For each band number, we computed the overall spectral universality correlation and compared to the null model. This sensitivity analysis tests the robustness of universality to discretization choices.

2.9 Rescaling Parameter Validation

To test whether spectral universality depends specifically on λ_2 rescaling or emerges more generally, we repeated the complete analysis using alternative rescaling parameters: λ_3 (third eigenvalue), λ_4 (fourth eigenvalue), λ_{mean} (mean of all non-zero eigenvalues), λ_{max} (maximum eigenvalue), and no rescaling at all (raw eigenvalues).

For each rescaling choice, we computed band-averaged profiles and inter-subject correlations, comparing universality metrics across all six approaches. This tests whether λ_2 is uniquely suited for revealing universality or whether the phenomenon is rescaling-independent.

2.10 Alternative Connectivity Methods

To verify that spectral universality is not an artifact of Pearson correlation-based connectivity, we repeated the analysis using two alternative functional connectivity measures: (1) Phase-Locking Value (PLV), which quantifies phase synchronization between electrode pairs and is sensitive to phase relationships independent of amplitude; and (2) Magnitude-Squared Coherence, which measures frequency-domain correlation and is sensitive to spectral power relationships.

For computational efficiency, we performed this analysis on a subset of 5 subjects. For each connectivity method, we computed connectivity matrices, Laplacian spectra, λ_2-rescaled eigenvalues, and spectral universality correlations. We also computed the coefficient of variation (CV) of λ_2 values across subjects for each method to assess relative variability.

3 Results

3.1 Spectral Universality in Resting State

Analysis of eyes-closed resting-state data revealed extraordinarily high spectral universality across subjects (Figure 1). After λ_2 rescaling, the five-band eigenvalue profiles showed mean inter-subject correlation of $r = 0.9852$, indicating that 98.52% of spectral structure variance is shared across individuals (Figure 1A). Individual band correlations ranged from $r = 0.9485$ (Band 5, highest frequencies) to $r = 0.9967$ (Band 4), with all bands exceeding $r = 0.94$ (Figure 2A).

The distribution of 45 pairwise correlations was tightly clustered around the mean, with standard deviation $\sigma = 0.0124$ and range 0.96-1.01 (Figure 2B). This narrow distribution indicates consistent universality across all subject pairs, not just high average correlation. Band 4 showed the highest universality ($r = 0.9967$, $\sigma = 0.0030$), explaining 99.67% of variance—among the highest cross-individual consistencies reported for any neural measure (Figure 2C).

The spectral gap λ_2 showed modest inter-subject variability (mean = 0.709, SD = 0.104, CV = 14.6%), with individual values ranging from 0.53 to 0.85 (Figure 3A,B). Despite this 60% range in absolute timescales, rescaling by individual λ_2 values revealed the underlying universal structure.

3.2 Null Model Comparison

Comparison to null models demonstrated that observed universality cannot arise by chance (Figure S1). The null distribution (1000 randomized realizations) yielded mean correlation $r_{\text{null}} = 0.0024$ with 95% confidence interval [-0.020, 0.028] (Figure S1A). The observed correlation $r = 0.9852$ is 410-fold greater than null expectation, with Z-score = 262.8 ($p < 0.000001$).

This effect size far exceeds conventional thresholds for extraordinary effects in psychology and neuroscience. The probability of observing such high correlation by chance is effectively zero, providing strong evidence for genuine spectral organization.

3.3 Band Sensitivity Analysis

Spectral universality proved robust to the number of bands used for discretization (Figure S2). Testing 3, 5, 7, and 10 bands yielded overall correlations of $r = 0.9891, 0.9852, 0.9798$, and 0.9352 respectively. All exceeded null expectations by ¿400-fold, confirming that universality is not an artifact of five-band discretization. The slight decrease with more bands likely reflects increased noise in finer-grained measurements rather than fundamental structure.

3.4 Multiple Comparisons Correction

All five frequency bands maintained statistical significance after Bonferroni correction (Table S1). With corrected threshold $\alpha = 0.01$, Z-scores ranged from 250.74 (Band 5) to 263.52 (Band 4), all yielding $p < 0.000001$. This demonstrates that spectral universality is not explained by multiple testing artifacts—each individual band independently shows extraordinary consistency across subjects.

The family-wise error rate (FWER) remained well-controlled at 4.9% (calculated as $1-(1-p)^5$ with $p < 0.000001$), far below the 5% threshold. This stringent statistical validation confirms that spectral universality represents genuine neural organization rather than statistical artifact.

3.5 Eyes-Open Control Analysis

Eyes-open resting state (R01) also exhibited high spectral universality, though slightly reduced compared to eyes-closed (Figure S3, Table S2). The overall correlation was $r = 0.9555$ (95.55% variance explained), representing a 3.1% decrease from eyes-closed ($r = 0.9852$).

Both conditions far exceeded null expectations ($r_{\text{null}} = 0.002$), confirming that universality is not specific to eyes-closed rest but generalizes across passive resting states. Per-band correlations ranged from 0.92-0.99 for eyes-open versus 0.95-1.00 for eyes-closed, with both conditions showing consistently high universality across all frequency bands.

The λ_2 distributions were statistically indistinguishable between conditions (eyes-open: mean = 0.717, SD = 0.080; eyes-closed: mean = 0.709, SD = 0.104; $t = 0.39, p = 0.71$), indicating that the spectral gap itself does not differ between states. The slight reduction in universality with eyes-open likely reflects increased variability from visual processing rather than fundamental differences in spectral organization.

3.6 Task-Induced Perturbation

Motor imagery task engagement substantially reduced spectral universality compared to resting state (Figure 1C, Figure 3C). Task-state correlation decreased to $r = 0.7351$, a reduction of

$\Delta r = -0.2502$ (25.4% decrease) from resting baseline. This reduction was consistent across all five frequency bands, with per-band correlations ranging from 0.73-0.75 during task versus 0.95-1.00 during rest.

Despite the reduction, task-state universality remained 300-fold higher than null expectation, suggesting that spectral organization persists during task engagement but is modulated by cognitive demands. The systematic reduction across all bands indicates a global perturbation from equilibrium rather than band-specific effects, consistent with external driving perturbing an otherwise universal equilibrium state.

3.7 Topology Independence

Spectral similarity showed no correlation with topological similarity between subjects (Figure 4; $\rho = 0.11, p = 0.34$). Despite substantial variation in individual connectivity patterns (connectivity similarity $r = 0.52$), rescaled spectral structures were nearly identical (spectral similarity $r = 0.99$).

This dissociation demonstrates that different wiring diagrams produce nearly identical spectral dynamics after λ_2 rescaling. Subjects with only 52% overlap in connectivity strength patterns nonetheless exhibit 99% spectral similarity, indicating that universality emerges from dynamical principles rather than shared anatomical constraints. This topology-independence strongly supports the interpretation that spectral universality reflects fundamental organizational principles of neural dynamics.

3.8 Rescaling-Independent Universality

Remarkably, all six rescaling approaches—$\lambda_2, \lambda_3, \lambda_4, \lambda_{\mathrm{mean}}, \lambda_{\mathrm{max}}$, and no rescaling—yielded identical spectral universality (Table S3; $r = 0.9852$ for all methods, Figure S4). Every rescaling parameter produced the same per-band correlations (Band 1: 0.9935, Band 2: 0.9941, Band 3: 0.9932, Band 4: 0.9967, Band 5: 0.9485), with differences below numerical precision.

This rescaling-independence reveals that spectral universality is more fundamental than specific theoretical predictions about λ_2. While PMIR theory motivated the use of λ_2 as the natural timescale for rescaling, the empirical data show that universality emerges at the signal level itself, independent of rescaling choices. Even raw, unrescaled eigenvalues exhibit the same universal structure when compared across subjects.

This finding strengthens rather than weakens the main conclusions: it demonstrates that spectral universality is an intrinsic property of resting brain dynamics, not an artifact of a particular mathematical transformation. The phenomenon transcends specific rescaling parameters, pointing to deeper organizational principles.

3.9 Connectivity Method Robustness

Spectral universality proved robust across different connectivity measurement approaches (Table S4, Figure S5). Phase-Locking Value (PLV) showed the highest universality ($r = 0.9974$, 99.74%

variance explained), followed by Pearson correlation ($r = 0.9902$, 99.02%), and magnitude-squared coherence ($r = 0.8538$, 85.38%). All three methods far exceeded null expectations ($r_{\text{null}} = 0.002$), demonstrating that universality is not an artifact of Pearson correlation.

The λ_2 coefficient of variation (CV) varied across methods: PLV showed highest variability (CV = 17.06%), Pearson intermediate (CV = 14.99%), and coherence lowest (CV = 7.84%). Despite these differences in λ_2 variability, all methods revealed high spectral universality, confirming robustness to measurement approach.

Per-band analysis revealed that phase-based methods (PLV, Pearson) showed consistently high universality across all five bands ($r > 0.95$), while frequency-domain coherence exhibited greater variation (Bands 1,2,4: $r > 0.97$; Bands 3,5: $r \approx 0.62 - 0.69$). This suggests that spectral universality is most robustly captured by phase relationships, though all methods reveal the underlying organization.

4 Discussion

4.1 Interpretation of Spectral Universality

The extraordinarily high spectral universality we observe ($r = 0.9852, p < 0.000001$) represents among the highest cross-individual neural consistencies reported in the literature [4, 5]. For comparison, typical test-retest reliabilities for fMRI connectivity measures range from $r = 0.3 - 0.7$, and cross-individual similarities rarely exceed $r = 0.8$. Our finding that 98.5% of spectral variance is shared across individuals suggests a fundamental organizational principle operating in resting brain networks.

This universality likely reflects equilibrium dynamics characteristic of self-organized systems. In equilibrium, systems settle into stable states determined by their governing equations rather than initial conditions or individual variations. The spectral structure we observe—organized into five distinct frequency bands with hierarchical timescales—may represent such an equilibrium configuration, where neural dynamics naturally organize into discrete temporal modes.

The independence from connectivity topology is particularly striking. That different anatomical wiring patterns produce identical spectral dynamics suggests that universality emerges from local interaction rules rather than global network architecture [6]. This parallels other universal phenomena in physics where microscopic details wash out at macroscopic scales, leaving only essential symmetries and conservation laws.

4.2 Rescaling Independence: Implications

The finding that spectral universality persists across all rescaling approaches has important theoretical implications. It suggests that the universal spectral structure is an intrinsic property of resting brain dynamics rather than an artifact of mathematical preprocessing. This stability across transformations indicates a deep organizing principle.

One interpretation is that the brain's functional connectivity networks possess intrinsic spectral organization that is revealed rather than created by rescaling. The nearly perfect correlation across subjects may reflect shared constraints from neural biophysics (e.g., refractory periods, synaptic timescales, oscillatory dynamics) that impose universal spectral structure regardless of individual anatomical variations.

The rescaling-independence also has practical implications: it means that spectral universality can be assessed without requiring specific theoretical commitments about which rescaling is "correct." This makes the measure more robust for clinical and research applications, as the same universality pattern emerges from multiple analytical approaches.

4.3 Clinical and Research Implications

Spectral universality provides a candidate biomarker for characterizing healthy brain function and potentially identifying deviations in neurological or psychiatric disorders. The extraordinarily high consistency in healthy individuals ($r > 0.98$) establishes a baseline against which patient populations could be compared.

Disorders affecting large-scale brain organization—such as schizophrenia, autism spectrum disorder, Alzheimer's disease, or epilepsy—might show reduced spectral universality relative to healthy controls. Such deviations could provide mechanistic insights into disease processes and potentially guide targeted interventions.

The rescaling-independence and methodological robustness make spectral universality particularly attractive for clinical translation, as it does not require specific analytical choices or theoretical commitments. The measure could be computed from standard EEG or MEG recordings using multiple approaches, all converging on the same underlying organization.

4.4 Limitations and Future Directions

Several limitations warrant discussion. First, our sample size (10 subjects for primary analysis, 5 for connectivity method comparison) is modest, though the extraordinarily high effect sizes and robust statistical significance provide strong evidence for the observed phenomena. Future work with larger samples could examine individual differences in detail and test generalization across populations.

Second, we analyzed only resting-state conditions with one task comparison. Systematic investigation of how different cognitive tasks, pharmacological manipulations, or clinical conditions affect spectral universality would provide deeper understanding of its functional significance and potential clinical utility.

Third, we used EEG data with relatively low spatial resolution (64 channels). Future work with high-density EEG, MEG, or fMRI could examine whether spectral universality holds at finer spatial scales and whether it shows regional variations. The combination of high temporal (EEG/MEG) and spatial (fMRI) resolution would be particularly valuable [10].

5 Conclusion

We have demonstrated extraordinarily high spectral universality in human resting-state brain networks, with 98.5% of spectral structure variance shared across individuals after λ_2 rescaling. This universality: (1) survives stringent statistical validation including Bonferroni correction; (2) generalizes across brain states (eyes-open and eyes-closed rest); (3) emerges identically across all rescaling parameters tested, revealing fundamental organization at the signal level; (4) proves robust across connectivity measurement methods; and (5) shows independence from individual connectivity topology, indicating emergence from dynamical rather than anatomical constraints.

These findings provide evidence that resting brain dynamics exist in a universal equilibrium characterized by spectral organization into discrete frequency bands. External task demands perturb this equilibrium, reducing universality while preserving structure. The phenomenon proves more fundamental than specific theoretical predictions, emerging robustly across analytical approaches.

Spectral universality may provide a principled framework for understanding healthy brain function and identifying deviations in neurological disorders. The rescaling-independence and methodological robustness make it particularly promising for clinical translation. Future work examining patient populations, computational mechanisms, and relationships to cognitive function will deepen understanding of this fundamental organizational principle in brain networks.

Acknowledgments

We thank the contributors to the PhysioNet EEG Motor Movement/Imagery Dataset for making their data publicly available. This work was conducted independently without institutional support.

Data and Code Availability

Data: PhysioNet EEG Motor Movement/Imagery Dataset (https://physionet.org/content/eegmmidb/)

Analysis code and processed data will be made available upon publication.

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